

Computer Vision Assessment Report

Task 1 — Basic Image Manipulation

What I built:

I loaded an image and performed three operations using OpenCV:

- I converted image to Grayscale, applied gaussian blur for smoothing, and used canny edge detection to extract edges

Challenges I faced:

- I faced challenges such as understanding how cv2 performs behind the scenes such as how each transformation works mathematically and tuning the parameters like kernel size, etc.

What I'd improve:

- For improvement I would add a slider to dynamically control blur strength and Canny thresholds using a UI.

Task 2 — Color Detector Using Webcam

What I built:

- I created a Python script that opens a webcam feed and shows the BGR color value of any pixel the user clicks, and I also added a close button for better usability.

Challenges I faced:

- I faced problems in handling mouse events and drawing a custom close button using OpenCV functions. I took help from google on sites like stack overflow on the possible error i got during the coding phases.

What I'd improve:

- Adding a color name detector using a pre-built color dictionary and a search algorithm would be a good option on the addition.

Conclusion:

This assessment helped me make better my understanding of basic computer vision techniques using OpenCV. With more time, I'd make the system interactive and add color classification using machine learning.

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